

# AICE 2009 Lab 4

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## Part 1

### Mainlobe Steering and Phase Terms

In the beampattern plot, the mainlobe points exactly in the direction of the steering angle. The steering vector phase term  $2\pi dn \sin(\theta)$  represents the expected spatial phase difference of an incoming wave arriving from angle  $\theta$  at the  $n$ -th microphone. By applying this phase term, the beamformer mathematically reverses the physical delays, aligning the signals from all microphones so they add up in phase for that specific direction.

### Effect of Number of Elements (N)

When comparing beampatterns for various element counts at a constant spacing, the array aperture becomes larger as the number of elements increases. This causes the mainlobe to become significantly narrower and sharper. Furthermore, the number of sidelobes increases, but they become narrower and their relative peak levels drop, which improves the array's ability to reject off-axis noise.

### Effect of Element Spacing (d)

Examining different element spacings reveals that additional strong lobes start appearing when the spacing reaches and exceeds one full wavelength. Spacing larger than a half-wavelength causes spatial aliasing. This means wavefronts arriving from completely different angles produce the exact same phase shifts across the array, making it impossible for the system to distinguish between the true source direction and the ambiguous directions.

### Tapering Windows

Using windowing functions like Hann and Hamming tapers the signal amplitudes at the edges of the array. The main benefit of this tapering is a reduction in the height of the sidelobes. However, the cost paid for this suppression is a noticeably wider mainlobe and reduced peak sharpness.

### Conventional vs. MVDR in a Two-Source Case

In the two-tone simulation with a  $20^\circ$  desired signal and a  $25^\circ$  interferer, conventional beamforming struggles to suppress the interferer because its mainlobe is too wide. The interferer falls within the mainlobe or a high sidelobe, and the conventional method lacks the ability to adaptively place a null. Conversely, MVDR is an adaptive beamformer that actively places a spatial null exactly at  $25^\circ$  to cancel the interferer. This drastically improves the Signal-to-Interference-plus-Noise Ratio (SINR). The conventional method yields an output SINR of -12.47 dB, whereas the MVDR method improves this to -0.13 dB.

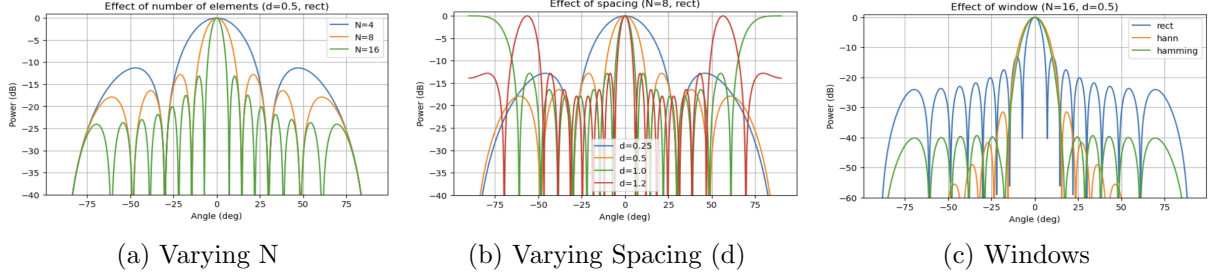


Figure 1: Beampattern comparisons for Part 1.

## Part 2

### Physical Meaning of the GCC-PHAT Peak

In the GCC-PHAT plot, the peak location (or lag) physically represents the Time Difference of Arrival (TDOA). It maps to the exact time delay it takes for the acoustic wavefront to travel the extra distance between the two microphones.

### Effect of SNR on GCC-PHAT

When the Signal-to-Noise Ratio (SNR) is lowered, such as from 20 dB to 0 dB, the prominent GCC-PHAT correlation peak becomes broader and less distinct. The general noise floor rises, and false peaks increase in amplitude. Consequently, the angle error increases because noise creates uncertainty in the exact peak location, sometimes causing false peaks to overtake the true delay peak entirely.

### DOA Estimates: GCC-PHAT vs. Delay-and-Sum Scan

As the source angle moves away from the broadside ( $0^\circ$ ) toward the endfire ( $\pm 90^\circ$ ), the estimation error noticeably increases. This is a geometric limitation of linear arrays; the time delay changes very little with respect to angle near the endfire due to the cosine relationship, making the system highly sensitive to small measurement errors.

Based on the 10 dB SNR outputs, the two-mic delay-and-sum scan gives significantly smaller errors overall. While both methods perform perfectly at broadside ( $0^\circ$ ), the GCC-PHAT implementation struggles severely at off-axis angles, producing errors up to  $139.7^\circ$  at a  $70^\circ$  true angle. This large discrepancy occurs because the GCC-PHAT lag estimates frequently flip signs compared to the true delay (for example, estimating  $+195.31\mu s$  instead of  $-201.99\mu s$ ). The delay-and-sum power scan correctly captures the spatial geometry and avoids these sign-flipping anomalies, restricting its maximum errors to just  $16.5^\circ$ .

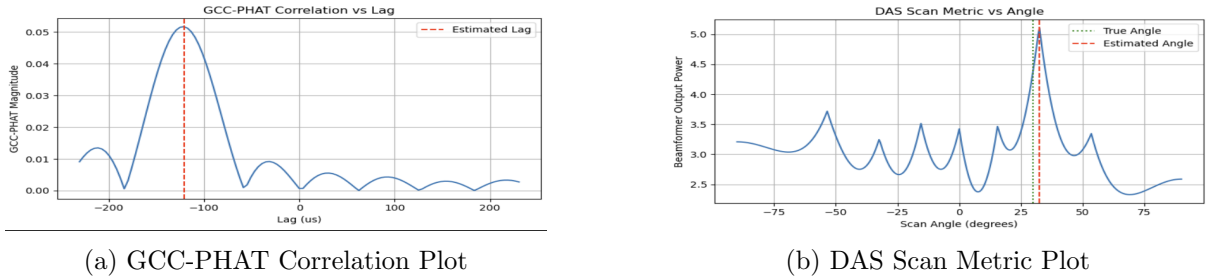


Figure 2: Comparison of GCC-PHAT and DAS Scan Metrics.